

Rajendran Prathibha Devkar

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Software engineer with 4+ years of production experience spanning full-stack systems (TypeScript, C#, Python, .NET) and applied ML (PyTorch, YOLOv8, TrOCR). MSCS candidate at UT Arlington. Published researcher with industry-proven instincts for performance and scale. Seeking roles at the intersection of applied AI and robust software engineering.

EXPERIENCE

Board Americas - QA / Software Developer Intern, Remote (USA)

May 2025 - Mar 2026

- Accelerated platform stability and release quality by delivering 40+ production fixes across an enterprise forecasting platform using Aurelia (TypeScript) and C#, directly reducing post-release hot fixes.
- Enhanced data integrity and visual trust in customer-facing analytics by resolving critical D3.js rendering defects involving fiscal/calendar year logic, axis precision mismatches, and sparse actual-versus-forecast plotting.
- Optimized high-volume workflows and UI consistency by re-engineering Workbench modules and resolving a favorites cache invalidation issue, eliminating stale session states and improving real-time data synchronization for concurrent users.
- Ensured 100% regression-free deployments by troubleshooting cross-layer issues spanning UI state and backend locking flows while partnering with senior engineers through peer code reviews.

Infosys Pvt. Ltd. - Technology Analyst Kerala, India

Oct 2019 - Nov 2022

- Improved user satisfaction by 40% by architecting a React Native curbside pickup application that streamlined end-to-end ordering and scheduling workflows.
- Scaled e-learning platform capacity to support 1,000+ concurrent users by developing and maintaining high-throughput RESTful APIs for session management.
- Reduced data retrieval latency by 40% by optimizing SQL query performance and re-engineering relational schema designs while maintaining 100% data integrity.
- Boosted application performance by 20% by executing JMeter load tests to simulate peak traffic, identifying and resolving architectural bottlenecks prior to deployment.
- Ensured high availability and 98% uptime by deploying services with PM2 to support multi-server rollouts and automated process management.
- Eliminated critical production crashes related to Calendar API integrations by resolving race conditions, significantly improving system reliability under heavy concurrent usage.

PROJECTS

Intelligent Handwritten Document Recognition (HTR) System

- Built an end-to-end handwritten document digitization pipeline that preserved layout semantics and exported structured Markdown and JSON using a hybrid synthetic + real annotated dataset.
- Trained a YOLOv8s layout detector (76.11% mAP@50) and integrated TrOCR-large for OCR (33% CER) after evaluating multiple engines and rejecting an overfit fine-tuned model.

COVID-19 Detection using Chest X-Rays

- Built a transfer-learning pipeline to classify chest X-ray images into COVID-19, pneumonia, and normal using the COVID-QU-Ex dataset (approx. 33.9K images).
- Used EfficientNet-B3 and ResNet-50 feature extractors with a regularized classifier, achieving 93.16% and 92.38% test accuracy, and deployed the model as a Streamlit web app for real-time inference.

EDUCATION

Master of Science in Computer Science,
University of Texas at Arlington, Texas

Aug 2024 - May 2026

Bachelor of Technology in Computer Science,
Dr. APJ Abdul Kalam Technological University, India

May 2019

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, C#, Java, C/C++

Frontend: Aurelia, React, React Native, HTML, CSS, D3.js, Bootstrap

ML / Data : NumPy, Pandas, PyTorch, TensorFlow, Keras, OpenCV, Matplotlib,

Cloud : AWS, Streamlit Cloud

AI / LLM: RAG, LLM Evaluation, Embeddings, Vector Databases, Prompt Engineering, LangChain, Semantic Kernel, Pinecone

Backend / APIs: .NET, Node.js, RESTful APIs

Databases: SQL Server, PostgreSQL, MySQL

DevOps / Tools: Azure DevOps, Git, Visual Studio, VS Code, Postman, PgAdmin, Jira

Testing / Quality: Regression testing, API testing, Performance testing (JMeter)

PUBLICATIONS

Daniel, M. S., John, N. P., **Devkar, R. Prathibha**, Abraham, R., & George, R. E. (2021). [Speaking Mouth System for Speech-Im-paired People Using Hand Gestures](#). *Advances in Intelligent Systems and Computing*, vol 1163. Springer, Singapore.

- Developed a webcam-based hand gesture recognition system that converts real-time gestures into text and speech using a CNN-based pipeline.
- Achieved up to 96% recognition accuracy using ROI-based preprocessing and grayscale/HSV image processing, and received KSCSTE INNOVATE research funding for the project.